Lab 3

Interfacing a Digital Sensor: Sonar

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Abstract

The purpose of this lab was to interface an HC-SR04 ultrasonic sensor with the DE1-SoC board and display the distance to an object in centimeters. LPM counters were used to trigger the sensor and measure its echo pulse in clock periods, which was converted to centimeters with an LPM divider and shown on the 7-segment displays using the display_8bits circuit from Lab 2. Echo widths measured on an ADALM 2000 oscilloscope at 10, 20, and 30 cm were compared with the displayed distance and a ruler.

Introduction

Sonar finds the distance to an object by sending out a sound pulse and timing how long the reflection takes to return. A pulse of at least 10 µs on the HC-SR04's TRIG input sends out an 8-cycle, 40 kHz burst. The ECHO output goes high when the burst is sent and low when the reflection is detected, so the echo pulse width equals the round-trip travel time. Since both signals are logic-level pulses, the sensor connects directly to the FPGA's GPIO pins. Distance is found by halving the echo width and multiplying by the speed of sound:

$$d = (t_{pw} / 2) \times v, \quad v = 343 \text{ m/s}$$

The FPGA measures time in cycles of the 50 MHz clock, which are 0.02 µs each. For the pre-lab, the time for sound to travel 1 cm one way is:

$$t = 0.01 \text{ m} / 343 \text{ m/s} = 29.15 \text{ µs}$$

$$29.15 \text{ µs} / 0.02 \text{ µs} = 1457.7 \approx 1458 \text{ clock periods}$$

Since the echo measures the round trip, each centimeter adds 2 × 1457.7 = 2915.5 clock periods, so a round-trip count N converts to distance as:

$$d \text{ (cm)} = (N \times 0.02 \text{ µs} \times 343 \text{ m/s} \times 100 \text{ cm/m}) / 2 = N / 2915.5 \approx N / 2915$$

Setup

This lab was completed in Quartus Prime Lite with an Intel DE1-SoC board. The HC-SR04 was connected to the GPIO 1 header (JP2) with M-F jumper wires and powered from the board's 3.3 V supply. The trigger and echo signals were also sent to a second set of pins so they could be viewed on channels 1 and 2 of an ADALM 2000 oscilloscope using Scopy. The connections are listed in Table 1. The display_8bits files from Lab 2 were reused, and the LPM_COUNTER, LPM_COMPARE, LPM_DIVIDE, and LPM_CONSTANT components were added from the IP Catalog.

Table 1: Connections between the DE1-SoC, the HC-SR04, and the ADALM 2000

Signal	FPGA Pin	JP2 Pin	Connected To
TRIG	PIN_AA21	2	HC-SR04 TRIG
TRIG_SCOPE	PIN_AB17	1	ADALM 2000 CH1
ECHO_IN	PIN_AC23	4	HC-SR04 ECHO
ECHO_SCOPE	PIN_AB21	3	ADALM 2000 CH2
+3.3 V	—	29	HC-SR04 Vcc
Ground	—	30	HC-SR04 GND
Ground	—	12	ADALM 2000 GND

Design

The circuit, shown in Figure 1, has three parts: a trigger section, an echo counter, and a conversion and display section.

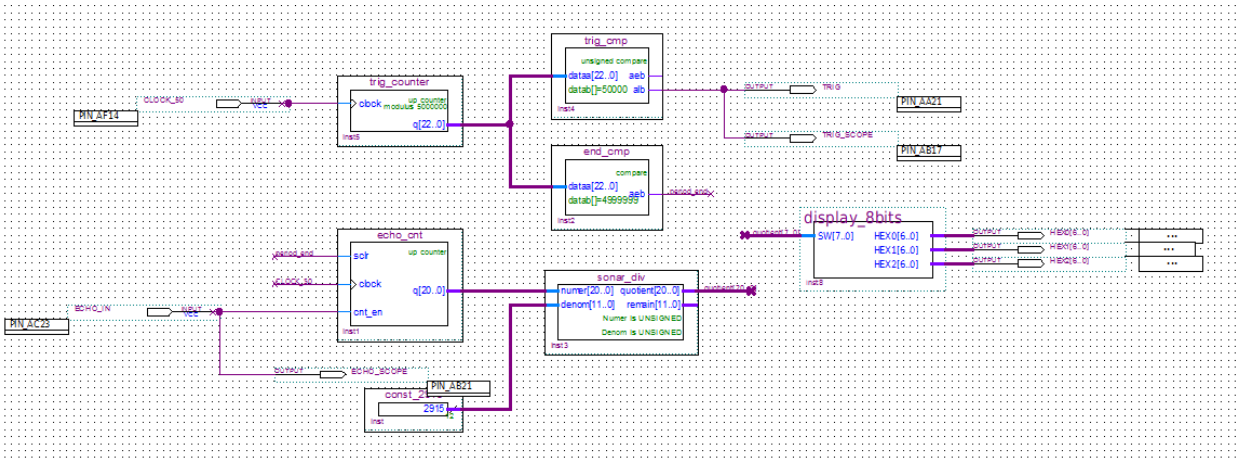


Figure 1: Schematic of the sonar circuit

Generating the Trigger Pulse

Trig_counter is a 23-bit LPM_COUNTER clocked by CLOCK_50 with a modulus of 5,000,000, so it rolls over every 100 ms and sets the measurement rate at 10 Hz. Its output feeds trig_cmp, an LPM_COMPARE whose alb output is high while the count is below 50,000. This creates a 1 ms trigger pulse at the start of each period, which is longer than the required 10 µs so it is easier to see on the oscilloscope. The alb output drives both TRIG and TRIG_SCOPE.

Measuring the Echo Pulse

Echo_cnt is a 21-bit LPM_COUNTER with ECHO_IN connected to its count enable, so it only counts while the echo is high and its final value is the echo width in clock periods. 21 bits is enough to hold the longest possible echo, about 36 ms (1,800,000 counts) when no object is detected. A second comparator, end_cmp, sets aeb high when trig_counter reaches its last count, 4,999,999. This signal, named period_end, drives the synchronous clear of echo_cnt so the count resets as each new period begins. ECHO_IN is also passed directly to ECHO_SCOPE for channel 2 of the oscilloscope.

Converting and Displaying the Distance

Following the pre-lab equation, the echo count is divided by a constant 2915 using sonar_div, an unsigned LPM_DIVIDE. The remainder is left unconnected, so the result is rounded down to the nearest centimeter. The lower 8 bits of the quotient, quotient[7..0], are sent to the display_8bits block from Lab 2, which limits the display to 0 to 255 cm. As in Lab 2, display7 outputs segment a on O[6], so the HEX pins were assigned in reverse order from the DE1-SoC pin table.

Results & Analysis

Debugging the Circuit

When first tested, the display showed a flickering 688 that did not respond to objects. The oscilloscope showed a correct trigger on channel 1, but channel 2 stayed constantly high, so the echo counter never stopped counting. The JP2 pins had been counted from the wrong end of the header, which reversed the sensor's power and damaged it. Connecting ECHO_IN to ground dropped channel 2 to 0 V, confirming that the circuit worked and the stuck-high echo came from the sensor. With ECHO_IN grounded, the display also showed an upside-down A instead of 0, which showed that the segment order was reversed. The HEX pin assignments were flipped to fix this, and the sensor was replaced.

Distance Measurements

Scopy was set to 1 ms/div and 1 V/div, triggering on the rising edge of channel 1. A box was held parallel to the sensor at 10, 20, and 30 cm, and at each distance the time cursors were placed on the edges of the echo pulse to measure its width, as shown in Figures 2 through 4. The distance was calculated using d = (t_pw / 2) × 343 m/s and compared with the board and ruler in Table 2.

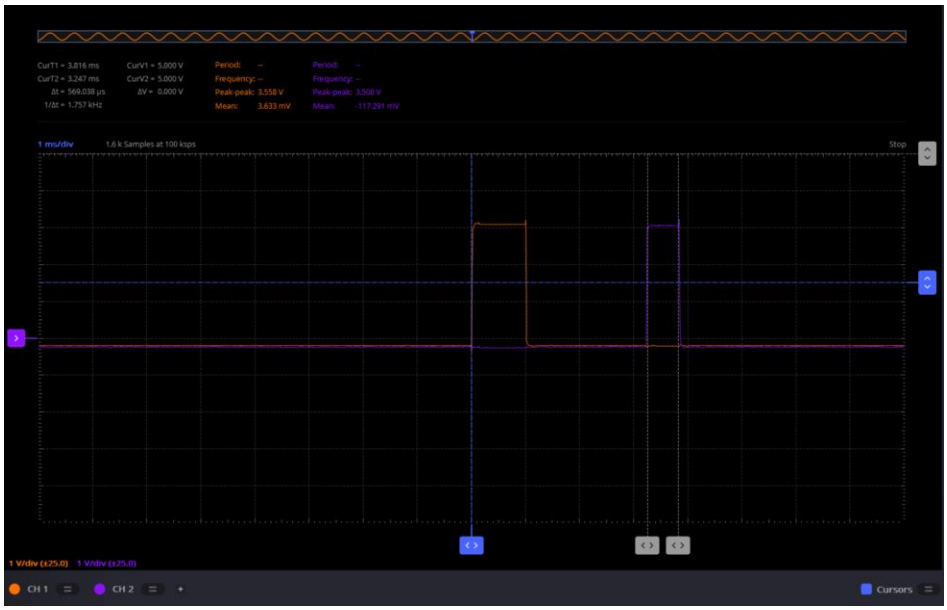


Figure 2: Trigger (CH1) and echo (CH2) at 10 cm, Δt = 569.0 µs

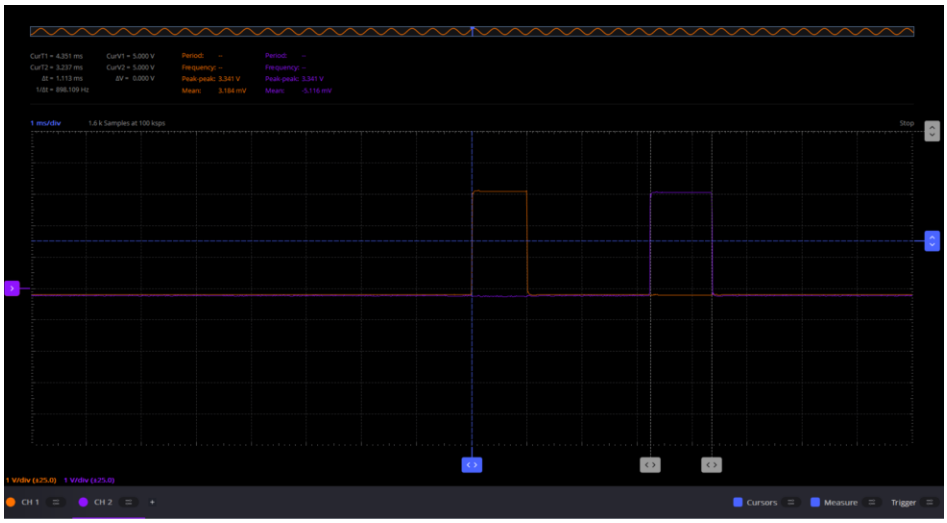


Figure 3: Trigger (CH1) and echo (CH2) at 20 cm, Δt = 1.113 ms

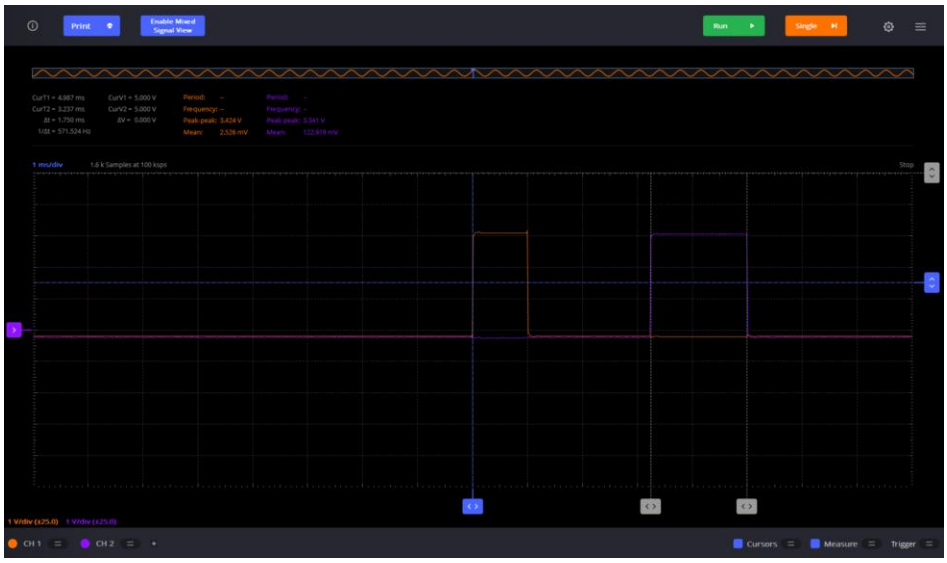


Figure 4: Trigger (CH1) and echo (CH2) at 30 cm, Δt = 1.750 ms

Table 2: Comparison of ruler, calculated, and displayed distances

Ruler (cm)	Echo Width	Calculated (cm)	Displayed (cm)	Calculated vs. Ruler Error
10	569.0 µs	9.76	10	2.4%
20	1.113 ms	19.09	20	4.6%
30	1.750 ms	30.01	30	0.0%

All calculated distances were within 1 cm of the ruler, and the board displayed 10, 20, and 30 cm. At 10 cm, the scope capture gives 9.76 cm, which the circuit would truncate to 9. The display reading of 10 was taken at a slightly different moment, and a few millimeters of hand

movement is enough to change the reading by 1 cm. The remaining error most likely came from holding the box by hand, lining the ruler up with the front of the sensor, and the speed of sound varying with room temperature. Figure 5 shows the sensor detecting the box at 30 cm.

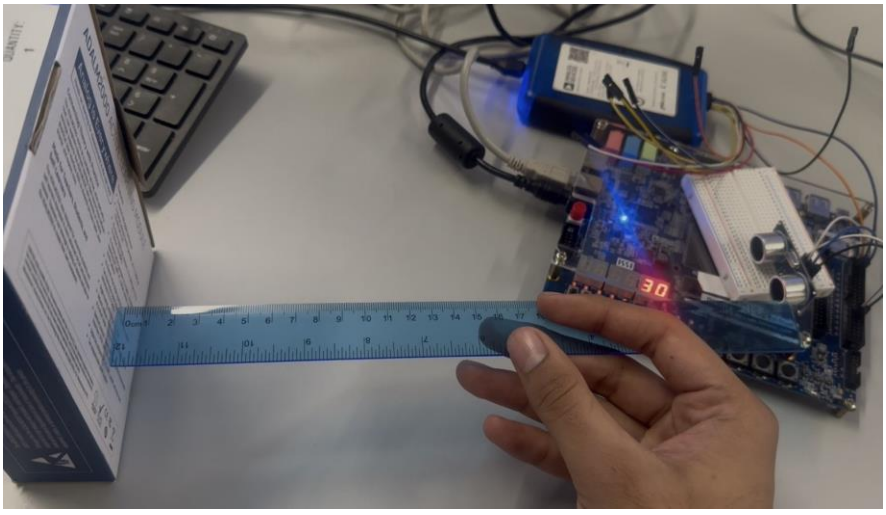


Figure 5: The sonar detecting a box at 30 cm, with the box, ruler, and display in view

Conclusion

The HC-SR04 was successfully interfaced with the DE1-SoC using counters to generate the trigger pulse and measure the echo in clock periods, which was then converted to centimeters and displayed using the display_8bits circuit from Lab 2. The calculated, displayed, and ruler distances all agreed within about 1 cm. The oscilloscope was especially useful for debugging, since viewing the trigger and echo signals directly separated sensor problems from circuit problems.

References

[1] “Lab 3, Interfacing a Digital Sensor: Sonar,” Northeastern University, EECE 2160, Fall 2026.

[2] “HC-SR04 Ultrasonic Sensor Module User Guide,” Handson Technology. [Online]. Available: https://www.handsontec.com/dataspecs/HC-SR04-Ultrasonic.pdf